Q3 — Build PRD Orchestrator

This is the complete written response for **Q3**. The presentation uses a concise summary and links back to this evidence.

Formal grading baseline: *v0.3.8 · n8n execution* `11901` *· run* `RUN-S2-11902-16e7090e`

Supplementary demo only — not the grading baseline: *n8n execution* `11958` *· run* `RUN-S2-11959-16e7090e`

Published view: *https://vjra.us/prd-genie.html#q3-build-architecture*

A governed multi-agent pipeline from source evidence to planning-ready delivery

Specialized agents accelerate documentation without replacing product judgment. PRD Orchestrator ingests six approved source documents, extracts and clarifies requirements, requires human approval, generates a corrected PRD, creates a dynamic Epic–Feature–User Story hierarchy, validates bidirectional traceability, and proposes non-blocking story sizes.

The v0.3.8 candidate uses the complete governed topology, restores sizing evaluation, removes duplicate semantic scoring, and captures fully loaded token and cost evidence.

Input-set evolution: three immutable discovery sources were followed by three human-authored clarification records dated August 7, 2026. Formal execution `11901` consumed the resulting six-document approved packet; decision IDs and bounded supersession preserve both original evidence and current governing state.

Architecture

Layer	Component	Responsibility	Evidence boundary
Source	Google Drive Inputs	Authoritative six-file runtime input	Drive ID, filename, normalized content, hash, retrieval time
Orchestration	n8n parent + seven stages + sizing	Run creation, structured handoffs, fail-closed routing	Run ID, workflow IDs, parent/child executions, contracts
Intelligence	OpenAI model calls	Extraction, gap analysis, PRD, story decomposition, sizing	Constrained outputs with exact evidence references
Human governance	Approval workflow	Business decisions, approved scope, citation dispositions	Signed approval lineage and terminal outcomes
Assurance	Code validators + Agreement Gate	Schema, sets, grounding, hierarchy, duplicates, orphans	Deterministic results and release authorization
Observability	Langfuse	Traces, evaluators, latency, tokens, cache, cost	Accepted parent/stage traces and evaluator evidence
Delivery	Google Drive Outputs	Validated Markdown and JSON artifact storage	Artifact identity, run association, delivery receipt
Evidence	GitHub · Obsidian · vjra.us	Version control, living inventory, submission walkthrough	Commits, snapshots, exported workflows, published views

The Gap Analyzer immediately follows Requirement Extraction to test the normalized, citation-linked contract before approval or PRD generation. This fail-fast boundary prevents unresolved gaps from becoming polished content; downstream validators separately check grounding, coverage, structure, and reconciliation.

Delivered capability

- Extraction: requirements, NFRs, stakeholders, constraints, and deadlines.
- Clarification: ambiguity and missing information remain explicit.
- PRD: corrected ten-section document with feature criteria.
- Delivery: 3 Epics · 7 Features · 11 Stories · 11 criteria.
- Validation: schema, grounding, reconciliation, and Agreement Gate.
- Export: seven validated run-specific artifacts.

- Gap analysis: missing information, conflicts, dependencies, and risks.
- Human-in-the-loop: approval before production generation.
- Traceability: 145/145 citations with forward and reverse lineage.
- Integrity: zero orphaned governed records.
- Sizing: 11/11 proposed sizes with confidence and rationale.

Formal baseline evaluator results

Stage	Code controls	Faithfulness	Hallucination	Additional result
Gap Analyzer	Passed	0.90	0.01	Gaps preserved without unsupported resolution
Human Approval	Passed	0.99	0.00	Approval lineage preserved
Production PRD	Passed	0.99	0.05	Required ten-section structure present
Story Breakdown	1.00	0.98	0.02	11 stories and criteria reconciled
Story Sizing	1.00	1.00	0.00	Reasonableness 1.00

T1–T10 test detailed, vague, contradictory, incomplete, persona, technical, dependency, and empty inputs. T11 validates PRD generation; T12 validates Epic and Story generation. Exact values are preserved, ambiguity is flagged without invention, acceptance criteria are preserved and mapped, and evidence is retained through n8n executions, Langfuse traces, and result files.

Accepted execution

- Parent execution: `11901`
- Run ID: `RUN-S2-11902-16e7090e`
- Duration: 2m 33.201s
- Agreement Gate: Release authorized
- Candidate status: Accepted v0.3.8 submission baseline

Cost and evaluation strategy

Cost component	Tokens	Cost	Operating interpretation
Pipeline generation	59,922	$0.228024	Core agent generation and transformation
41 evaluator calls	485,545	$1.246385	Deterministic controls are free; semantic evaluation drives loaded cost
Fully loaded total	545,467	$1.474409	Accepted candidate baseline for approximately one governed run per week

At one fully evaluated run per user per week, the modeled average is approximately **$0.21 per user per day** and **$6.38 per user per month**.

Cost control strategy: execute code controls first, remove duplicate semantic evaluators, apply stage-specific LLM judges only where needed, monitor cached and total tokens, retain full evaluation for candidate acceptance, and use lighter routine monitoring only after governance approval.

Q3 rubric coverage

Criterion	Points	Evidence
Architecture, orchestration, tool rationale	10	System diagram, logical flow, technical layers, selection rationale
Core capabilities end to end	12	Extraction through validated export
Extended capability	8	Gap analysis, human approval, traceability, sizing
Observability connected	5	n8n evidence and Langfuse code/LLM results
Baseline dataset tested	5	T1–T12 ground truth, outputs, traces, documented results
Cost and evaluation strategy	5	Loaded tokens, cost, thresholds, optimization strategy
Total	**45**	Fully represented

Q3 conclusion: The solution's principal contribution is not unrestricted document generation. It is the combination of specialized agents, evidence grounding, visible uncertainty, human approval, bidirectional traceability, deterministic validation, semantic evaluation, and fail-closed release control.